

# Model Notes

## Endogenous Education Choice in Segmented Search Markets

(Variant: skill-biased  $P_S$ , narrow deep set)

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**Scope.** This document contains the full mathematical description of the model: primitives, value functions, equilibrium conditions, solution algorithms, and transition dynamics. It serves as the companion reference for the paper and the codebase. **Changes relative to the baseline**

**(March 15) draft.** This variant differs from the baseline draft in four respects:

1. **Skill-biased skilled productivity.** The skilled productivity shifter is no longer a scalar  $P_S$ ; instead, it is an increasing function of worker type  $x$ , parameterised as the density of a one-parameter Beta distribution with shape parameter  $\gamma_{P_S} \geq 1$ . Concretely,

$$P_S(x) = \gamma_{P_S} x^{\gamma_{P_S}-1}, \quad \pi_S(x, p) = x P_S(x) p = \gamma_{P_S} x^{\gamma_{P_S}} p.$$

The scalar  $P_S$  is replaced by  $\gamma_{P_S}$  in the parameter classification.

2. **Narrowed deep parameter set.** The set of deep (estimation-invariant) structural parameters has been narrowed to  $\{a_\ell, b_\ell, b_U, b_T, b_S\}$ . The training cost coefficient  $c$ , the matching technology parameters  $(\eta_U, \eta_S, \mu_U, \mu_S)$ , the bargaining weights  $(\beta_U, \beta_S)$ , and the on-the-job search flow cost  $\sigma_S$ —previously treated as deep—are now re-estimated within each window. This reflects the possibility that crises differ in training incentives, matching, and wage-setting technology, not only in their aggregate state. Whether technology actually shifts within a crisis pair is then a testable empirical question rather than a modelling assumption.
3. **Training completion rate reclassified.** The training completion rate  $\phi$  has been moved from calibrated/fixed to pre-estimated from data, since it is computed directly from NSC/IPEDS Universe enrollment and NCES median time-to-degree rather than set externally.
4. **Notation simplified: no regime argument on parameters.** In previous drafts every regime-specific parameter (and many value functions) carried a  $(z)$  argument. The model itself is a single stationary equilibrium; the multi-regime structure is an estimation device for splitting the data into windows and is not part of the model description. All  $(z)$  arguments have been dropped from parameters, value functions, distributions, cutoffs, and densities, except in the transition-dynamics section, where a brief pre-switch / post-switch distinction is unavoidable. When a regime change is described, the relevant parameters simply jump from one set of values to another at the switch date.

All other primitives, value functions, equilibrium conditions, algorithms, and transition-dynamics objects are unchanged relative to the baseline draft. In particular, skilled workers do not search the unskilled market, and unskilled free entry uses only the unskilled-unemployed pool.

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# 1 Global environment

Time is continuous. All agents discount at rate  $r > 0$ .

## 1.1 Sectoral productivity shifters

There are two sector-specific productivity shifters  $P_U > 0$  and  $P_S(\cdot)$ . Whereas  $P_U$  is a constant scalar, the skilled productivity shifter is an *increasing* function of worker type  $x$ , parameterised as the density of a one-parameter Beta distribution:

$$P_S(x) = \gamma_{P_S} x^{\gamma_{P_S}-1}, \quad x \in [0, 1], \quad \gamma_{P_S} \geq 1, \quad (1)$$

i.e.  $P_S(\cdot) \sim \text{Beta}(\gamma_{P_S}, 1)$ . The single parameter  $\gamma_{P_S}$  replaces the old constant  $P_S$ . Setting  $\gamma_{P_S} = 1$  recovers a flat  $P_S(x) = 1$  (skill-neutral);  $\gamma_{P_S} > 1$  generates a strict skill premium.

## 1.2 Demographic turnover

Workers exit the economy at Poisson rate  $\nu > 0$  and are replaced one-for-one by newborn untrained workers with density  $\ell(x)$ . This delivers a stationary population composition.

## 1.3 Stationarity

The model admits a stationary equilibrium (steady state) in which the cross-sectional distribution of workers across states is time-invariant. The transition-dynamics section (Section 11) studies the path generated by an unexpected, permanent change in some parameters at a date  $T$ .

# 2 Parameter classification

Parameters fall into four groups, following the multi-regime estimation strategy described in the paper plan. The classification identifies which parameters are externally set, which are estimated once on the baseline window and held fixed, and which are re-estimated within each window. All parameters appear without a regime argument in the equations below; classification governs how each one is *estimated*, not the model itself.

## 2.1 Calibrated / fixed

| Symbol | Name          | Source / calibration                 |
|--------|---------------|--------------------------------------|
| $r$    | Discount rate | 4% annual real rate (risk-free rate) |

The discount rate is set externally and held fixed throughout all estimation stages. It is calibrated rather than pre-estimated because no single moment identifies  $r$  independently of the flow payoffs ( $b_U, b_T, b_S$ ) and the productivity shifters; including  $r$  in the SMM objective would render the Hessian near-singular.

## 2.2 Pre-estimated from data

| Symbol | Name                      | Source / method                                                                                                     |
|--------|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| $\phi$ | Training completion rate  | NSC/IPEDS Universe enrollment + NCES median time-to-degree, enrollment-weighted across 4-year and 2-year programmes |
| $\nu$  | Demographic turnover rate | CPS Basic Monthly LF→NILF exit hazard (ages 16–64) over the pre-FC baseline window                                  |

Both rates are pre-estimated from external micro-data outside the SMM objective. The training completion rate is computed as

$$\hat{\phi} = \frac{E_{4\text{yr}}/d_{4\text{yr}} + E_{2\text{yr}}/d_{2\text{yr}}}{E_{4\text{yr}} + E_{2\text{yr}}},$$

where  $E_k$  is the average IPEDS Universe enrollment in  $k$ -year institutions over the sample and  $d_k$  is the corresponding NCES median time-to-degree (49 months for 4-year bachelor's; 37 months for 2-year associate's). Pre-estimating  $\phi$  from external data avoids the near-collinearity with the training cutoff  $\bar{x}$  and the type distribution  $(a_\ell, b_\ell)$  that arises when both are identified jointly from the training share and flow. The turnover rate  $\hat{\nu}$  equals the average monthly LF→NILF exit rate for the working-age population (16–64) over the pre-FC baseline window. Pre-estimating  $\nu$  avoids the collinearity with  $r$  that would arise if both were included in the SMM objective, while grounding the value in observed transition data rather than a life-table assumption.

### 2.3 Deep structural parameters

| Symbol           | Name                     | Interpretation             |
|------------------|--------------------------|----------------------------|
| $a_\ell, b_\ell$ | Type distribution (Beta) | Worker heterogeneity       |
| $b_U$            | Unskilled UI flow value  | Institutional UI policy    |
| $b_T$            | Training flow value      | Training stipend / support |
| $b_S$            | Skilled UI flow value    | Institutional UI policy    |

These parameters govern the *fundamentals* of the labour market that we assume are invariant across crisis episodes: the underlying distribution of innate worker types and institutional flow payoffs that reflect government UI / training policies which change on slower timescales than the business cycle. All deep parameters are estimated once on the pre-FC baseline window and held fixed across all subsequent windows.

### 2.4 Regime-specific parameters (re-estimated within each window)

| Symbol                 | Name                         | Interpretation               |
|------------------------|------------------------------|------------------------------|
| $c$                    | Training cost coefficient    | Cost of education            |
| $P_U, \gamma_{P_S}$    | Productivity shifters        | Aggregate state              |
| $k_U, k_S$             | Vacancy costs                | Hiring conditions            |
| $\lambda_U, \lambda_S$ | Shock arrival rates          | Match-destruction risk       |
| $\alpha_U$             | Unskilled damage shape       | Damage-shock distribution    |
| $a_\Gamma, b_\Gamma$   | Skilled offer shape          | Offer / quality distribution |
| $\xi_S$                | Exogenous skilled separation | Layoff rate                  |
| $\eta_U, \eta_S$       | Matching elasticities        | Matching-function curvature  |
| $\mu_U, \mu_S$         | Matching efficiencies        | Matching-function level      |
| $\beta_U, \beta_S$     | Bargaining weights           | Wage-setting institutions    |
| $\sigma_S$             | OJS flow cost                | On-the-job search cost       |

These parameters capture both the *aggregate state* of the economy (productivity, separation hazards, vacancy costs), the *matching and wage-setting technology*, and the *training technology*. Allowing the matching parameters  $(\eta_j, \mu_j)$ , the bargaining weights  $(\beta_j)$ , the OJS cost  $\sigma_S$ , and the training cost coefficient  $c$  to be re-estimated within each window is essential for testing whether crises that look superficially similar in fact differ in the underlying technology of the labour market or of education. A pure demand shock should leave  $(c, \eta_j, \mu_j, \beta_j, \sigma_S)$  approximately constant across the base→crisis pair within a single crisis (e.g. FC), whereas a supply / reallocation shock that re-curves the Beveridge curve need not. Distinguishing these two cases empirically is the central identifying use of the multi-regime structure.

### 3 Workers, firms, and types

#### 3.1 Workers

Each worker has an innate type  $x \in \mathcal{X} = [0, 1]$ , fixed over time. Types are distributed according to CDF  $L$  with density  $\ell$  ( $x \sim \text{Beta}(a_\ell, b_\ell)$ ).

**Market access via training (persistent eligibility).** Workers can participate in either the *unskilled* market  $U$  (untrained status) or the *skilled* market  $S$  (trained status). Training changes *market access only*; it does not change innate type  $x$ .

**Training decision at unskilled unemployment.** Training can be initiated *only* by untrained workers who are in unskilled unemployment. The optimal policy is characterised by a cutoff  $\bar{x} \in [0, 1]$ : upon entering unskilled unemployment, types  $x > \bar{x}$  initiate training, while types  $x \leq \bar{x}$  remain in unskilled search.

#### 3.2 Firms

There are two labour markets. Both markets have homogeneous firms.

### 4 Training technology

**Eligibility.** Only untrained workers in unskilled unemployment may initiate training.

**Cost.** Initiating training requires a one-time cost  $c(x)$  with  $c'(x) < 0$ . In applications,

$$c(x) = (1 - x) e^{(c-x)}.$$

Higher-ability workers face lower training costs. The parameter  $c$  enters through exponentiation so that the optimiser searches over  $c \in \mathbb{R}$  while the cost coefficient  $e^c > 0$  is guaranteed positive.

**Completion.** Training completes at Poisson rate  $\phi > 0$ . Upon completion, the worker becomes trained and enters skilled unemployment.

**Flow payoffs.** Let  $b_U$ ,  $b_T$ , and  $b_S$  denote flow payoffs while unskilled unemployed, in training, and skilled unemployed, respectively.

**Demographic turnover (stationarity device).** Workers exit the economy at Poisson rate  $\nu > 0$  and are replaced one-for-one by newborn untrained workers with density  $\ell(x)$ .

### 5 Production technologies

**Unskilled production.** An unskilled match between worker type  $x$  and a homogeneous firm with match-specific efficiency  $p \in (0, 1]$  produces

$$\pi_U(x, p) = P_U x p.$$

**Skilled production.** A skilled match between worker type  $x$  and a homogeneous firm with idiosyncratic quality  $p \in (0, 1]$  produces

$$\pi_S(x, p) = x P_S(x) p = \gamma_{P_S} x^{\gamma_{P_S}} p, \tag{2}$$

so the per-unit-quality factor is  $x P_S(x) = \gamma_{P_S} x^{\gamma_{P_S}}$ , strictly increasing in  $x$  when  $\gamma_{P_S} > 1$ .

## 6 Unskilled labour market

### 6.1 Matching

Let  $\mathcal{U}_U$  and  $\mathcal{V}_U$  denote the masses of unskilled unemployed and unskilled vacancies. Market tightness is

$$\theta_U = \frac{\mathcal{V}_U}{\mathcal{U}_U}.$$

Matching follows a Cobb–Douglas function with parameters  $\mu_U > 0$  and  $\eta_U \in (0, 1)$ :

$$M_U(\mathcal{U}_U, \mathcal{V}_U) = \mu_U \mathcal{U}_U^{\eta_U} \mathcal{V}_U^{1-\eta_U},$$

implying vacancy filling and job-finding rates

$$q_U(\theta_U) = \mu_U \theta_U^{-\eta_U}, \quad \theta_U q_U(\theta_U) = \mu_U \theta_U^{1-\eta_U}.$$

### 6.2 Match productivity and shocks

New matches are created at frontier efficiency  $p_0 = 1$ . Idiosyncratic damage shocks arrive at Poisson rate  $\lambda_U > 0$ . Upon a shock, post-shock efficiency is

$$p' = e^{-d}, \quad d \sim \text{Exp}(\alpha_U),$$

equivalently

$$p' \sim \text{Beta}(\alpha_U, 1), \quad G(p) = p^{\alpha_U}, \quad p \in [0, 1].$$

A match is endogenously destroyed whenever  $p' < p^*(x)$  for a reservation threshold  $p^*(x) \in (0, 1)$ .

### 6.3 Value functions and training policy

Let:

- $U_U(x)$  be the value of untrained unskilled unemployment,
- $U_U^{\text{search}}(x)$  the value of unskilled search,
- $T(x)$  the value of training,
- $E_U(x, p)$  the worker value of unskilled employment at efficiency  $p$ ,
- $J_U(x, p)$  the firm value of a filled unskilled job,
- $V_U$  the value of an unskilled vacancy.

**Household values and training policy.** An untrained unemployed worker chooses between unskilled search and initiating training:

$$U_U(x) = \max\{U_U^{\text{search}}(x), U_U^{\text{train}}(x)\}, \quad U_U^{\text{train}}(x) := -c(x) + T(x).$$

Training completes at Poisson rate  $\phi$  and leads to skilled unemployment. The training value satisfies

$$(r + \phi + \nu) T(x) = b_T + \phi U_S(x). \quad (3)$$

Unskilled search yields flow payoff  $b_U$  and contacts vacancies at rate  $\theta_U q_U(\theta_U)$ :

$$(r + \nu) U_U^{\text{search}}(x) = b_U + \theta_U q_U(\theta_U) (E_U(x, 1) - U_U^{\text{search}}(x)). \quad (4)$$

Define the net training gain

$$\Delta_T(x) := -c(x) + T(x) - U_U^{\text{search}}(x).$$

Under monotone incentives, there exists a cutoff  $\bar{x} \in [0, 1]$  such that

$$\tau(x) = \mathbf{1}\{x > \bar{x}\}, \quad U_U^{\text{search}}(\bar{x}) = -c(\bar{x}) + T(\bar{x}).$$

## 6.4 Match values, bargaining, and endogenous separation

A new unskilled match starts at frontier efficiency  $p_0 = 1$ . Idiosyncratic damage shocks arrive at rate  $\lambda_U$  and redraw efficiency  $p'$  from  $G$  on  $[0, 1]$ . For  $p \in (0, 1]$ , worker and firm values satisfy

$$(r + \nu + \lambda_U) E_U(x, p) = w_U(x, p) + \lambda_U \left[ U_U(x) G(p^*(x)) + \int_{p^*(x)}^1 E_U(x, p') dG(p') \right], \quad (5)$$

$$(r + \nu + \lambda_U) J_U(x, p) = P_U x p - w_U(x, p) + \lambda_U \int_{p^*(x)}^1 J_U(x, p') dG(p'). \quad (6)$$

Wages are determined by Nash bargaining with worker weight  $\beta_U \in (0, 1)$ :

$$\beta_U J_U(x, p) = (1 - \beta_U)(E_U(x, p) - U_U(x)). \quad (7)$$

Endogenous separation is characterised by a reservation threshold  $p^*(x) \in (0, 1)$  satisfying

$$J_U(x, p^*(x)) = 0, \quad J_U(x, p) \geq 0 \iff p \geq p^*(x). \quad (8)$$

## 6.5 Pointwise update for unskilled search values

Although the training option implies that the bargaining outside value is the full unemployment value  $U_U(x) = \max\{U_U^{\text{search}}(x), U_U^{\text{train}}(x)\}$ , the search value itself admits a convenient pointwise representation. Rearranging (4) gives

$$(r + \nu + \theta_U q_U(\theta_U)) U_U^{\text{search}}(x) = b_U + \theta_U q_U(\theta_U) E_U(x, 1). \quad (9)$$

Using Nash bargaining at entry efficiency  $p_0 = 1$ ,

$$E_U(x, 1) = U_U(x) + \frac{\beta_U}{1 - \beta_U} J_U(x, 1),$$

we obtain the pointwise update

$$U_U^{\text{search}}(x) = \frac{b_U + \theta_U q_U(\theta_U) \left( U_U(x) + \frac{\beta_U}{1 - \beta_U} J_U(x, 1) \right)}{r + \nu + \theta_U q_U(\theta_U)}. \quad (10)$$

Under the cutoff policy  $\tau(x) = \mathbf{1}\{x > \bar{x}\}$ ,

$$U_U(x) = \begin{cases} U_U^{\text{search}}(x), & x \leq \bar{x}, \\ U_U^{\text{train}}(x), & x > \bar{x}, \end{cases}$$

so (10) becomes a simple piecewise update. In particular, on the search region  $x \leq \bar{x}$  it reduces to

$$(r + \nu) U_U^{\text{search}}(x) = b_U + \frac{\beta_U}{1 - \beta_U} \theta_U q_U(\theta_U) J_U(x, 1).$$

## 6.6 Unskilled match surplus (wage-free equation)

Define the total match surplus as worker net gain plus firm value:

$$S_U(x, p) \equiv (E_U(x, p) - U_U(x)) + J_U(x, p).$$

Equivalently,  $E_U(x, p) + J_U(x, p) = S_U(x, p) + U_U(x)$ . Adding the worker and firm HJBs (5)–(6) and substituting  $E_U + J_U = S_U + U_U$ :

$$(r + \nu + \lambda_U) S_U(x, p) = P_U x p - (r + \nu) U_U(x) + \lambda_U \int_{p^*(x)}^1 S_U(x, p') dG(p'). \quad (11)$$



## 6.7 Wage schedule (unskilled)

Given the surplus split  $J_U = (1 - \beta_U)S_U$  and  $E_U - U_U = \beta_U S_U$ , wages are recovered from the firm HJB (6). Multiplying the surplus equation (11) by  $(1 - \beta_U)$  and comparing with the firm HJB gives

$$w_U(x, p) = \beta_U P_U x p + (1 - \beta_U)(r + \nu) U_U(x). \quad (12)$$

The wage is linear in match quality  $p$ : the worker captures share  $\beta_U$  of flow output  $P_U x p$ , plus a flow option value  $(1 - \beta_U)(r + \nu)U_U(x)$  reflecting the firm's outside value when the worker exits (at rate  $\nu$ ) or the match is destroyed and the worker returns to  $U_U$ .

## 6.8 Vacancies, free entry, and market tightness

Let  $V_U$  denote the value of an unskilled vacancy. Posting a vacancy costs  $k_U$  per unit time and yields contacts at rate  $q_U(\theta_U)$ . Let  $u_U(x)$  denote the density of untrained unskilled unemployed by type and define

$$\mathcal{U}_U = \int_0^1 u_U(x) dx.$$

A vacancy meets type  $x$  with probability  $u_U(x)/\mathcal{U}_U$  and creates a match at  $p_0 = 1$ , yielding value  $J_U(x, 1)$ . The vacancy HJB is

$$rV_U = -k_U + q_U(\theta_U) \left( \frac{1}{\mathcal{U}_U} \int_0^1 J_U(x, 1) u_U(x) dx - V_U \right). \quad (13)$$

Free entry implies

$$V_U = 0 \quad \implies \quad q_U(\theta_U) = \frac{k_U \mathcal{U}_U}{\int_0^1 J_U(x, 1) u_U(x) dx}, \quad (14)$$

which pins down market tightness  $\theta_U$  given the endogenous composition of unemployed workers.

## 6.9 Reservation rule

Setting  $S_U(x, p^*) = 0$  in the surplus equation (11):

$$p^*(x) = \frac{(r + \nu) U_U(x) - \lambda_U \int_{p^*}^1 S_U(x, p') dG(p')}{P_U x}. \quad (15)$$

## 6.10 Stationary cross-sectional distributions (unskilled)

The cross-sectional distribution of *untrained* workers across states is pinned down by flow balance. Let  $u_U(x)$  denote the density of untrained workers in unskilled unemployment,  $t(x)$  the density of workers in training, and define the untrained-segment mass by type:

$$m_U(x) \equiv \ell(x) - m_S(x),$$

where  $m_S(x) = \frac{\phi}{\nu} t(x)$  is the trained mass (in stationarity). The untrained employed density is the residual:

$$e_U(x) := m_U(x) - u_U(x) - t(x).$$

**Training.** At equilibrium the training distribution is stationary:

$$0 = \tau(x) u_U(x) - (\phi + \nu) t(x), \quad \implies \quad t(x) = \frac{\tau(x)}{\phi + \nu} u_U(x). \quad (16)$$

**Untrained unemployment.** Untrained unemployment receives inflow from demographic replacement and job destruction, and loses mass due to job finding, training entry, and demographic exit. The density flow balance is

$$\left(\theta_U q_U(\theta_U) + \tau(x) + \nu\right) u_U(x) = \nu \ell(x) + \lambda_U G(p^*(x)) e_U(x), \quad (17)$$

where  $e_U(x) = m_U(x) - u_U(x) - t(x)$ .

## 7 Unskilled-block equilibrium and solution algorithm

An *unskilled-block equilibrium* is a collection

$$\{\theta_U, U_U^{\text{search}}(\cdot), T(\cdot), U_U(\cdot), E_U(\cdot, \cdot), J_U(\cdot, \cdot), p^*(\cdot), u_U(\cdot), t(\cdot)\}$$

such that:

- (E1) **Household values and training.**  $T(x)$  satisfies (3);  $U_U^{\text{search}}(x)$  satisfies (4) (equivalently (10)); and unemployment values satisfy  $U_U(x) = \max\{U_U^{\text{search}}(x), -c(x) + T(x)\}$ ,  $\tau(x) = \mathbf{1}\{-c(x) + T(x) \geq U_U^{\text{search}}(x)\}$ .
- (E2) **Match values and wages.** Worker and firm match values satisfy (5)–(6); wages satisfy Nash bargaining (7). (Equivalently,  $w_U(x, p)$  is recovered from (12).)
- (E3) **Endogenous separation.** The reservation threshold  $p^*(x)$  satisfies (8).
- (E4) **Vacancy creation (free entry).** Market tightness  $\theta_U$  satisfies free entry (14).
- (E5) **Stationary cross-sectional distributions.** Densities satisfy the flow-balance conditions (16) and (17). Employment of untrained workers is the residual mass  $e_U(x) = m_U(x) - u_U(x) - t(x)$ , where  $m_U(x) = \ell(x) - m_S(x)$ .

### 7.1 Solution algorithm (unskilled side)

For computation, it is convenient to organise the unskilled block as a two-layer fixed point. The *inner loop* solves the value functions  $(U_U^{\text{search}}, U_U, T, E_U, J_U)$  given conjectures for market tightness  $\theta_U$  and the reservation rule  $p^*(\cdot)$ . The *outer loop* updates the policy objects  $(\tau(\cdot), p^*(\cdot), \theta_U)$  and the stationary composition  $(u_U, t)$  until all equilibrium conditions hold jointly.

**Outer-loop unknowns.** We iterate on

$$\left(\theta_U^{(n)}, p^{*,(n)}(\cdot), \tau^{(n)}(\cdot), u_U^{(n)}(\cdot), t^{(n)}(\cdot)\right).$$

**Step 1. Initialise (outer loop).** Choose initial guesses  $\theta_U^{(0)} > 0$  and a feasible reservation rule  $p^{*,(0)}(x) \in (0, 1)$  (e.g. constant in  $x$ ). Initialise a training policy  $\tau^{(0)}(x)$  (or cutoff  $\bar{x}^{(0)}$ ) and stationary densities  $u_U^{(0)}(x)$  and  $t^{(0)}(x)$  (e.g. proportional to  $\ell$ ).

**Step 2. Inner value-function iteration (given  $\theta_U^{(n)}$  and  $p^{*,(n)}$ ).** Holding  $(\theta_U^{(n)}, p^{*,(n)}(\cdot))$  fixed, iterate jointly on  $(U_U^{\text{search}}, U_U, T, E_U, J_U)$  until convergence. One convenient inner cycle is:

- **Step 2.1. Training value.** Update  $T^{(m+1)}(x)$  from (3), treating  $U_S(\cdot)$  as given by the skilled block.
- **Step 2.2. Unemployment value and training decision.** Given  $T^{(m+1)}$  and the current search value  $U_U^{\text{search},(m)}$ , update the full outside value

$$U_U^{(m+1)}(x) = \max\{U_U^{\text{search},(m)}(x), -c(x) + T^{(m+1)}(x)\}$$

and the implied training policy

$$\tau^{(m+1)}(x) = \mathbf{1}\{-c(x) + T^{(m+1)}(x) \geq U_U^{\text{search},(m)}(x)\}.$$

- **Step 2.3. Match values.** Given  $U_U^{(m+1)}(x)$  and the fixed reservation rule  $p^{*,(n)}(\cdot)$ , solve the coupled system (5)–(6) together with bargaining (7) to obtain  $E_U^{(m+1)}(x, p)$  and  $J_U^{(m+1)}(x, p)$ .
- **Step 2.4. Search value.** Update  $U_U^{\text{search},(m+1)}(x)$  from the HJB (4):

$$(r + \nu) U_U^{\text{search},(m+1)}(x) = b_U + \theta_U^{(n)} q_U(\theta_U^{(n)})(E_U^{(m+1)}(x, 1) - U_U^{\text{search},(m+1)}(x)).$$

Stop the inner loop when the sup norm of updates to  $(U_U^{\text{search}}, U_U, E_U, J_U)$  is below tolerance.

**Step 3. Update stationary distributions (given the converged policy and hazards).**

Using the converged training policy  $\tau^{(n+1)}(\cdot)$  from the inner loop and the current reservation rule  $p^{*,(n)}(\cdot)$ , update the stationary densities  $(u_U^{(n+1)}, t^{(n+1)})$  by solving (16)–(17) pointwise in  $x$  (with  $e_U(x) = m_U(x) - u_U(x) - t(x)$ ).

**Step 4. Update reservation rule  $p^*$  (outer update).** Setting  $S_U(x, p^*) = 0$  in the closed-form surplus expression yields the direct formula (15). Anderson acceleration (rank-1) is applied to the vector  $\{p^{*,(n+1)}(x)\}_x$  before writing it back to cache.

**Step 5. Update tightness  $\theta_U$  by free entry (outer update).** Given the updated composition  $u_U^{(n+1)}(\cdot)$  and the converged frontier surplus  $J_U(x, 1)$ , update  $\theta_U$  using free entry (14):

$$q_U(\theta_U^{(n+1)}) = \frac{k_U \mathcal{U}_U^{(n+1)}}{\int_0^1 J_U(x, 1) u_U^{(n+1)}(x) dx},$$

and invert  $q_U(\theta) = \mu_U \theta^{-\eta_U}$  to obtain  $\theta_U^{(n+1)}$ . Use Anderson acceleration when updating tightness, to reduce fluctuations.

**Step 6. Convergence (outer loop).** Stop if

$$\max\left\{|\theta_U^{(n+1)} - \theta_U^{(n)}|, \|p^{*,(n+1)}(\cdot) - p^{*,(n)}(\cdot)\|_\infty, \|u_U^{(n+1)}(\cdot) - u_U^{(n)}(\cdot)\|_\infty\right\} < \text{tol}.$$

Otherwise set  $n \leftarrow n + 1$  and return to Step 2.

**Remark 1** (Two-block model). *In the full model, Step 2 takes  $U_S(\cdot)$  as given from the skilled block, while the skilled block takes the trained mass  $m_S(\cdot) = \frac{\phi}{\nu} t(\cdot)$  (in steady state) as given from the unskilled block. A global equilibrium is obtained by iterating on the two blocks until these link variables are consistent.*

## 8 Skilled labour market

This section adapts the Pissarides (2000) Chapter 4 framework (“Labour Turnover and On-the-Job Search”) to our segmented environment with worker heterogeneity  $x \in [0, 1]$ . The skilled block features (i) idiosyncratic match quality  $p \in [0, 1]$  that is redrawn by shocks, (ii) endogenous separation via a cutoff  $p_S^*(x)$ , and (iii) on-the-job search (OJS) as a discrete choice with a flow search cost  $\sigma_S$ .

### 8.1 Primitives and market objects

**Trained population.** Only trained workers participate in the skilled market. Let  $m_S(x)$  be the density of trained workers of type  $x$ . In stationarity,

$$m_S(x) = \frac{\phi}{\nu} t(x),$$

as implied by training completion and demographic turnover.

**Production.** A skilled match between worker type  $x$  and idiosyncratic quality  $p \in [0, 1]$  produces  $\pi_S(x, p) = \gamma_{P_S} x^{\gamma_{P_S}} p$  (see (2)).

**Separations and quality shocks.** A filled skilled match is destroyed exogenously at Poisson rate  $\xi_S > 0$ . In addition, match quality is hit by shocks at Poisson rate  $\lambda_S > 0$ . Upon a shock, quality is redrawn i.i.d. from  $\Gamma$  on  $[0, 1]$  with density  $\gamma(\cdot)$ . A match is endogenously destroyed whenever the post-shock draw is below a reservation cutoff  $p_S^*(x) \in (0, 1)$ .

**Interpretation of  $\Gamma$ .** The distribution  $\Gamma$  is the *match-quality distribution* in the skilled market. It governs both (i) initial match formation (offers to unemployed workers) and (ii) quality redraws after idiosyncratic shocks. Unlike the unskilled market, skilled matches are not born at a fixed frontier; instead, match quality is drawn from  $\Gamma$  at formation and after shocks. We parameterise  $\Gamma = \text{Beta}(a_\Gamma, b_\Gamma)$ .

**OJS (discrete choice).** An employed skilled worker chooses whether to search while employed. Searching incurs a flow cost  $\sigma_S \geq 0$  (paid by the worker). If searching, the worker receives outside offers at the same contact rate as an unemployed job seeker,  $\theta_S q_S(\theta_S)$ . Each offer is a draw  $\tilde{p} \sim \Gamma$ . The worker quits to the outside job iff the offer is strictly better (job ladder),  $\tilde{p} > p$ .

**Demographic turnover.** All workers exit at Poisson rate  $\nu > 0$ .

## 8.2 Matching and tightness

Let  $u_S(x)$  denote the density of skilled unemployed by type, and let  $e_S(x, p)$  denote the density of employed trained workers by  $(x, p)$ . Define the OJS policy  $s^*(x, p) \in \{0, 1\}$  (defined below), and the aggregate mass of skilled job seekers as

$$\tilde{U}_S \equiv \underbrace{\int_0^1 u_S(x) dx}_{U_S} + \underbrace{\int_0^1 \int_0^1 s^*(x, p) e_S(x, p) dp dx}_{\mathcal{E}_S^{\text{search}}}.$$

Let  $\mathcal{V}_S$  be the mass of skilled vacancies and define tightness as

$$\theta_S = \frac{\mathcal{V}_S}{\tilde{U}_S}.$$

Matching is Cobb–Douglas in vacancies and job seekers, with parameters  $\mu_S > 0$  and  $\eta_S \in (0, 1)$ :

$$M_S(\tilde{U}_S, \mathcal{V}_S) = \mu_S \tilde{U}_S^{\eta_S} \mathcal{V}_S^{1-\eta_S},$$

implying the vacancy contact (filling) rate

$$q_S(\theta_S) = \mu_S \theta_S^{-\eta_S},$$

and the contact rate for any job seeker (unemployed or employed searcher)  $\theta_S q_S(\theta_S)$ .

## 8.3 Value functions, bargaining, and two wage schedules

Let:

- $U_S(x)$  be the value of skilled unemployment;
- $E_S^0(x, p)$  and  $E_S^1(x, p)$  be worker values in a job of quality  $p$  without and with OJS;
- $J_S^0(x, p)$  and  $J_S^1(x, p)$  be the corresponding firm values;
- $V_S$  be the value of a skilled vacancy.

Define the worker's continuation value in a job of quality  $p$  as

$$E_S(x, p) \equiv \max\{E_S^0(x, p), E_S^1(x, p)\}, \quad J_S(x, p) \equiv J_S^{s^*(x, p)}(x, p).$$

**Surplus and Nash bargaining.** For each search regime  $s \in \{0, 1\}$  define match surplus

$$S_S^s(x, p) \equiv (E_S^s(x, p) - U_S(x)) + J_S^s(x, p).$$

Nash bargaining with worker weight  $\beta_S \in (0, 1)$  implies

$$E_S^s(x, p) - U_S(x) = \beta_S S_S^s(x, p), \quad J_S^s(x, p) = (1 - \beta_S) S_S^s(x, p), \quad s \in \{0, 1\}. \quad (18)$$

This delivers two wage schedules  $w_S^0(x, p)$  and  $w_S^1(x, p)$  because the continuation values differ across  $s$  (search changes quit hazards and introduces the search cost).

**Unemployed worker.** A skilled unemployed worker meets a vacancy at rate  $\theta_S q_S(\theta_S)$  and draws  $\tilde{p} \sim \Gamma$ . The worker accepts iff the resulting match has nonnegative surplus (equivalently  $\tilde{p} \geq p_S^*(x)$ ). New matches start with the worker free to choose whether to search or not, i.e. the acceptance value is  $E_S(x, \tilde{p})$ . The unemployment HJB is

$$(r + \nu) U_S(x) = b_S + \theta_S q_S(\theta_S) \int_{p_S^*(x)}^1 (E_S(x, \tilde{p}) - U_S(x)) d\Gamma(\tilde{p}). \quad (19)$$

Write  $\kappa_S \equiv \theta_S q_S(\theta_S)$ .

**Employed worker, no search ( $s = 0$ ).** With no search, the only events are demographic exit  $\nu$ , exogenous separation  $\xi_S$ , and quality shocks  $\lambda_S$ . The worker HJB is

$$\begin{aligned} (r + \nu + \xi_S + \lambda_S) E_S^0(x, p) &= w_S^0(x, p) + \xi_S U_S(x) \\ &+ \lambda_S \left[ U_S(x) \Gamma(p_S^*(x)) + \int_{p_S^*(x)}^1 E_S(x, p') d\Gamma(p') \right]. \end{aligned} \quad (20)$$

**Employed worker, search ( $s = 1$ ).** Searching costs  $\sigma_S$  per unit time and generates outside contacts at rate  $\kappa_S$ . Upon an offer  $\tilde{p}$ , the worker quits iff  $\tilde{p} > p$  and the offered match is feasible. Hence

$$\begin{aligned} (r + \nu + \xi_S + \lambda_S) E_S^1(x, p) &= (w_S^1(x, p) - \sigma_S) + \xi_S U_S(x) \\ &+ \lambda_S \left[ U_S(x) \Gamma(p_S^*(x)) + \int_{p_S^*(x)}^1 E_S(x, p') d\Gamma(p') \right] \\ &+ \kappa_S \int_{\max\{p, p_S^*(x)\}}^1 (E_S(x, \tilde{p}) - E_S^1(x, p)) d\Gamma(\tilde{p}). \end{aligned} \quad (21)$$

**Firm values.** A filled job yields flow profit  $\gamma_{P_S} x^{\gamma_{P_S}} p - w_S^s(x, p)$ . With no search ( $s = 0$ ), there is no quit hazard. With OJS ( $s = 1$ ), the job is lost at rate  $\kappa_S(1 - \Gamma(p))$  because any offer above  $p$  triggers a quit. Thus:

$$\begin{aligned} (r + \nu + \xi_S + \lambda_S) J_S^0(x, p) &= \gamma_{P_S} x^{\gamma_{P_S}} p - w_S^0(x, p) \\ &+ \lambda_S \int_{p_S^*(x)}^1 J_S(x, p') d\Gamma(p'), \end{aligned} \quad (22)$$

$$\begin{aligned} (r + \nu + \xi_S + \lambda_S + \kappa_S(1 - \Gamma(p))) J_S^1(x, p) &= \gamma_{P_S} x^{\gamma_{P_S}} p - w_S^1(x, p) \\ &+ \lambda_S \int_{p_S^*(x)}^1 J_S(x, p') d\Gamma(p'). \end{aligned} \quad (23)$$

## 8.4 Policy rules: endogenous separation and OJS cutoff

**Endogenous separation cutoff.** Define overall surplus as  $S_S(x, p) \equiv \max\{S_S^0(x, p), S_S^1(x, p)\}$ . Endogenous separation is characterised by a cutoff  $p_S^*(x)$  satisfying

$$S_S(x, p_S^*(x)) = 0, \quad S_S(x, p) \geq 0 \iff p \geq p_S^*(x). \quad (24)$$

**OJS cutoff (search vs no-search).** The worker searches iff it is optimal:

$$s^*(x, p) = \mathbf{1}\{E_S^1(x, p) \geq E_S^0(x, p)\}.$$

Under monotonicity in  $p$ , there exists an OJS cutoff  $p_S^{\text{oj}}(x) \geq p_S^*(x)$  such that

$$E_S^1(x, p) \geq E_S^0(x, p) \iff p < p_S^{\text{oj}}(x), \quad E_S^1(x, p_S^{\text{oj}}(x)) = E_S^0(x, p_S^{\text{oj}}(x)). \quad (25)$$

Workers in low-quality matches search (the ladder gain exceeds  $\sigma_S$ ); workers in high-quality matches do not.

## 8.5 Surplus and Wages

**Step 1: useful identity.** Because  $E_S(x, p) = U_S(x) + \beta_S S_S(x, p)$  and  $J_S(x, p) = (1 - \beta_S) S_S(x, p)$  under the optimal regime at  $(x, p)$ , we have

$$E_S(x, p) + J_S(x, p) = U_S(x) + S_S(x, p).$$

Hence, for any lower bound  $\underline{p} \geq p_S^*(x)$ ,

$$\int_{\underline{p}}^1 (E_S(x, p') + J_S(x, p')) d\Gamma(p') = (1 - \Gamma(\underline{p})) U_S(x) + \int_{\underline{p}}^1 S_S(x, p') d\Gamma(p').$$

**Step 2: eliminate wages.** Adding the worker and firm HJBs in each search regime  $s$  and subtracting the unemployment HJB eliminates wages, leaving an HJB for  $S_S^s$ .

**Surplus with no search ( $s = 0$ ).**

$$\begin{aligned} (r + \nu + \xi_S + \lambda_S) S_S^0(x, p) &= \gamma_{P_S} x^{\gamma_{P_S}} p - b_S - (r + \nu) U_S(x) \\ &\quad + (\lambda_S - \beta_S \kappa_S) \int_{p_S^*(x)}^1 S_S(x, p') d\Gamma(p'). \end{aligned} \quad (26)$$

**Surplus with on-the-job search ( $s = 1$ ).**

$$\begin{aligned} (r + \nu + \xi_S + \lambda_S + \kappa_S(1 - \Gamma(p))) S_S^1(x, p) &= \gamma_{P_S} x^{\gamma_{P_S}} p - b_S - (r + \nu) U_S(x) - \sigma_S \\ &\quad + \lambda_S \int_{p_S^*(x)}^1 S_S(x, p') d\Gamma(p') - \beta_S \kappa_S \int_{p_S^*(x)}^p S_S(x, p') d\Gamma(p'). \end{aligned} \quad (27)$$

## 8.6 Wage schedules

**No OJS ( $s = 0$ ).** Multiplying (26) by  $(1 - \beta_S)$  and comparing with the firm HJB (22):

$$w_S^0(x, p) = \beta_S \gamma_{P_S} x^{\gamma_{P_S}} p + (1 - \beta_S) b_S + \beta_S(1 - \beta_S) \kappa_S \int_{p_S^*(x)}^1 S_S(x, p') d\Gamma(p'). \quad (28)$$

The wage is linear in  $p$  for fixed  $x$ : the third term is an  $x$ -specific constant that reflects the unemployment-contact option available to the worker (weight  $\beta_S$ ) and surrendered to the firm (weight  $1 - \beta_S$ ) when the match is valued under no search.

**OJS ( $s = 1$ ).** The same procedure applied to (27) and (23):

$$w_S^1(x, p) = \beta_S \gamma_{P_S} x^{\gamma_{P_S}} p + (1 - \beta_S)(b_S + \sigma_S) + \beta_S(1 - \beta_S) \kappa_S \int_{p_S^*(x)}^p S_S(x, p') d\Gamma(p'). \quad (29)$$

**OJS wage premium.** Comparing (29) and (28), the OJS wage premium is

$$w_S^1(x, p) - w_S^0(x, p) = -(1 - \beta_S) \sigma_S - \beta_S(1 - \beta_S) \kappa_S \int_p^1 S_S(x, p') d\Gamma(p'). \quad (30)$$

The premium is *non-positive*: the search cost  $(1 - \beta_S)\sigma_S$  is borne entirely by the worker (Nash shifts it from  $w_S^1$  relative to  $w_S^0$ ), while the ladder option term  $-\beta_S(1 - \beta_S)\kappa_S \int_p^1 S_S d\Gamma \leq 0$  reflects the firm's compensation by a wage discount proportional to the expected capital gain it would lose to a poaching offer. Both terms vanish as  $p \rightarrow 1$  (no further upward ladder) and are largest near  $p_S^*(x)$ .

## 8.7 Vacancies, poaching, and free entry

Posting a skilled vacancy costs  $k_S > 0$  per unit time. A vacancy meets a *random job seeker* at rate  $q_S(\theta_S)$ . Since job seekers include unemployed and employed searchers, the expected value of a contact depends on the composition of seekers. Define the density of job seekers (unemployed + employed searchers) as

$$\tilde{u}_S(x) \equiv u_S(x) + \int_0^1 s^*(x, p) e_S(x, p) dp, \quad \tilde{\mathcal{U}}_S = \int_0^1 \tilde{u}_S(x) dx.$$

Conditional on a contact, the worker type is drawn from  $\tilde{u}_S(\cdot)$ , and:

- if the contacted seeker is unemployed, the match quality draw is  $\tilde{p} \sim \Gamma$  and the match is formed iff  $\tilde{p} \geq p_S^*(x)$ , yielding value  $J_S(x, \tilde{p})$ ;
- if the contacted seeker is an employed searcher currently at quality  $p$ , the offer is  $\tilde{p} \sim \Gamma$  and the worker accepts (quits) iff  $\tilde{p} > p$  and  $\tilde{p} \geq p_S^*(x)$ , yielding value  $J_S(x, \tilde{p})$ .

Thus the vacancy HJB is

$$rV_S = -k_S + q_S(\theta_S)(\mathcal{J}_S - V_S), \quad (31)$$

where the expected surplus from a random contact is

$$\begin{aligned} \mathcal{J}_S \equiv & \underbrace{\frac{1}{\tilde{\mathcal{U}}_S} \int_0^1 u_S(x) \int_{p_S^*(x)}^1 J_S(x, \tilde{p}) d\Gamma(\tilde{p}) dx}_{\text{meeting unemployed}} \\ & + \underbrace{\int_0^1 \int_0^1 s^*(x, p) e_S(x, p) \int_{\max\{p, p_S^*(x)\}}^1 J_S(x, \tilde{p}) d\Gamma(\tilde{p}) dp dx}_{\text{poaching employed searchers}}. \end{aligned} \quad (32)$$

Free entry implies  $V_S = 0$ , hence

$$q_S(\theta_S) \mathcal{J}_S = k_S. \quad (33)$$

Given  $(u_S, e_S, s^*)$ , this pins down  $\theta_S$  by inverting  $q_S$ .

## 8.8 Stationary cross-sectional distributions (trained workers)

Let  $u_S(x)$  be the density of skilled unemployed and  $e_S(x, p)$  the density of employed trained workers at quality  $p$ . Accounting within the trained population:

$$m_S(x) = u_S(x) + \int_{p_S^*(x)}^1 e_S(x, p) dp.$$

**Endogenous destruction from quality shocks.** Quality shocks arrive at rate  $\lambda_S$  and redraw  $p' \sim \Gamma$ . A match is destroyed if  $p' < p_S^*(x)$ , so the endogenous destruction hazard is

$$\delta_S^{\text{end}}(x) \equiv \lambda_S \Gamma(p_S^*(x)).$$

**Job-to-job quit hazard under OJS.** If the worker searches ( $s^*(x, p) = 1$ ), outside offers arrive at rate  $\kappa_S$  and the worker quits iff  $\tilde{p} > p$ , with probability  $1 - \Gamma(p)$ . Hence

$$\delta_S^{\text{jj}}(x, p) \equiv s^*(x, p) \kappa_S (1 - \Gamma(p)).$$

**Unemployment balance.** For each  $x$ :

$$(\nu + \kappa_S(1 - \Gamma(p_S^*(x)))) u_S(x) = \phi t(x) + \int_{p_S^*(x)}^1 (\xi_S + \delta_S^{\text{end}}(x)) e_S(x, p) dp. \quad (34)$$

**Employment-by-quality balance.** For each  $x$  and each  $p \geq p_S^*(x)$ , the density  $e_S(x, p)$  satisfies a steady-state balance: outflows from the state  $(x, p)$  equal inflows into  $(x, p)$ . Outflows are due to demographic exit, exogenous separation, quality shocks, and job-to-job quits. Inflows come from three sources: (i) hiring from unemployment into quality  $p$  (offer density  $\gamma(p)$ ), (ii) quality shocks from any employed state into  $p$  (redraw density  $\gamma(p)$ ), and (iii) job-to-job upgrades from worse jobs  $p' < p$  when those workers search and draw  $p$ .

$$\begin{aligned} & (\nu + \xi_S + \lambda_S + \delta_S^{\text{jj}}(x, p)) e_S(x, p) \\ &= \underbrace{\kappa_S u_S(x) \gamma(p)}_{\text{hiring from unemployment into } p} + \underbrace{\lambda_S \gamma(p) \int_{p_S^*(x)}^1 e_S(x, p') dp'}_{\text{quality shocks into } p} \\ &+ \underbrace{\kappa_S \gamma(p) \int_{p_S^*(x)}^p s^*(x, p') e_S(x, p') dp'}_{\text{job-to-job inflow from worse jobs}}. \end{aligned} \quad (35)$$

**Remark 2** (Implementation). On a grid  $p_1 < \dots < p_{N_p}$ , equation (35) becomes a linear system in the unknown vector  $\{e_S(x, p_i)\}_{i=1}^{N_p}$  for each  $x$ : the quit hazard contributes to the diagonal; quality shocks add a rank-one inflow term proportional to  $\gamma(p_i)$ ; and the job-ladder inflow term is lower-triangular because it depends only on  $\{e_S(x, p_j)\}_{j < i}$  in the OJS region ( $s^* = 1$ ).

## 9 Skilled-block equilibrium and nested fixed-point algorithm

Take the trained population density  $m_S(\cdot)$  as given from the unskilled block. A *skilled-block equilibrium* is a collection

$$\{\theta_S, U_S(\cdot), E_S^0(\cdot, \cdot), E_S^1(\cdot, \cdot), J_S^0(\cdot, \cdot), J_S^1(\cdot, \cdot), w_S^0(\cdot, \cdot), w_S^1(\cdot, \cdot), \\ p_S^*(\cdot), p_S^{\text{oj}}(\cdot), s^*(\cdot, \cdot), u_S(\cdot), e_S(\cdot, \cdot)\}$$

such that:

- (S1) **Worker values.** Unemployment satisfies (19). Employed values satisfy (20)–(21), with  $E_S(x, p) \equiv \max\{E_S^0(x, p), E_S^1(x, p)\}$ .
- (S2) **Firm values.** Filled-job values satisfy (22)–(23), with  $J_S(x, p) \equiv J_S^{s^*(x, p)}(x, p)$ .
- (S3) **OJS policy.** The worker's search decision is optimal:  $s^*(x, p) = \mathbf{1}\{E_S^1(x, p) \geq E_S^0(x, p)\}$ , summarised by a cutoff  $p_S^{\text{oj}}(x)$  satisfying (25).
- (S4) **Endogenous separation.** The reservation cutoff  $p_S^*(x)$  satisfies (24).



- (S5) **Wage determination.** Wages satisfy Nash bargaining (18). (Equivalently,  $w_S^s$  are recovered from the worker HJBs (20)–(21) and checked/adjusted to enforce (18) pointwise.)
- (S6) **Tightness and free entry (Chapter 4 congestion).** Tightness is defined using job seekers (unemployed plus employed searchers) and free entry holds:  $q_S(\theta_S) \mathcal{J}_S = k_S$ .
- (S7) **Stationary distributions.** Given  $(\theta_S, p_S^*, p_S^{\text{oj}}, s^*)$ , the stationary densities  $(u_S, e_S)$  satisfy (34)–(35) and the accounting identity  $m_S(x) = u_S(x) + \int_{p_S^*(x)}^1 e_S(x, p) dp$ .

## 9.1 Nested fixed-point algorithm (skilled side)

For computation, it is convenient to organise the skilled block as a nested fixed point. The *inner loop* solves the value functions (and wage schedules) given outer objects. The *outer loop* updates  $(\theta_S, p_S^*, p_S^{\text{oj}})$  and the stationary composition until all equilibrium conditions hold jointly.

**Outer-loop unknowns.** Iterate on

$$\left( \theta_S^{(n)}, p_S^{*,(n)}(\cdot), p_S^{\text{oj},(n)}(\cdot) \right).$$

**Step 1. Initialise (outer loop).** Choose initial guesses  $\theta_S^{(0)} > 0$  and feasible cutoffs  $p_S^{*,(0)}(x) \in (0, 1)$  and  $p_S^{\text{oj},(0)}(x) \in [p_S^{*,(0)}(x), 1]$ . Initialise stationary densities  $(u_S^{(0)}, e_S^{(0)})$  consistent with  $m_S(\cdot)$  (e.g. set  $u_S^{(0)} = \kappa m_S$  and allocate the remainder of mass to the top  $p$  grid point).

**Step 2. Inner loop: surplus iteration (given outer objects).** Holding  $(\theta_S^{(n)}, p_S^{*,(n)}, p_S^{\text{oj},(n)})$  fixed, iterate on  $(U_S, S_S^0, S_S^1)$  until convergence. Let  $\kappa_S = \theta_S^{(n)} q_S(\theta_S^{(n)})$  throughout. One inner cycle for each  $x$ :

- **Step 2.1. Tail integral.** Let  $j_0$  be the first grid index with  $p_{j_0} \geq p_S^{*,(n)}(x)$ . Compute the tail integral using the *previous* iteration's surplus:  
 $\mathcal{I}^{(m)}(x) = \int_{p_S^{*,(n)}(x)}^1 S_S^{\text{max},(m)}(x, p') d\Gamma(p')$ , where  $S_S^{\text{max},(m)} = \max\{S_S^{0,(m)}, S_S^{1,(m)}\}$ . This is approximated by accumulating the weighted sum  $\sum_{k \geq j_0} S_S^{\text{max},(m)}(x, p_k) w_\Gamma(p_k)$  in a backward pass over the  $p$ -grid.
- **Step 2.2. Unemployment value (closed form).**

$$U_S^{(m+1)}(x) = \frac{b_S + \kappa_S \beta_S \mathcal{I}^{(m)}(x)}{r + \nu}.$$

- **Step 2.3. Surplus on  $p$ -grid (closed-form evaluation).** For each grid point  $p_j$ :
  - If  $j < j_0$  (infeasible region):  $S_S^{0,(m+1)} = S_S^{1,(m+1)} = 0$ .
  - If  $j \geq j_0$ :

$$S_S^{0,(m+1)}(x, p_j) = \frac{\gamma_{P_S} x^{\gamma_{P_S}} p_j - (r + \nu) U_S^{(m+1)} + \lambda_S \mathcal{I}^{(m)}}{r + \nu + \xi_S + \lambda_S},$$

$$S_S^{1,(m+1)}(x, p_j) = \frac{\gamma_{P_S} x^{\gamma_{P_S}} p_j - (r + \nu) U_S^{(m+1)} - \sigma_S + \lambda_S \mathcal{I}^{(m)} + \kappa_S \beta_S \int_{p_j}^1 S_S^{\text{max},(m)} d\Gamma}{r + \nu + \xi_S + \lambda_S + \kappa_S(1 - \Gamma(p_j))}.$$

The numerator integral  $\int_{p_j}^1 S_S^{\text{max},(m)} d\Gamma$  is available from Step 2.1.

- **Step 2.4. Nash split.** Recover employed values and firm values via  $E_S^{s,(m+1)} = U_S^{(m+1)} + \beta_S S_S^{s,(m+1)}$ ,  $J_S^{s,(m+1)} = (1 - \beta_S) S_S^{s,(m+1)}$ ,  $s \in \{0, 1\}$ . Wages are not computed explicitly. Stop when  $\sup_{x,p} \max\{|S_S^{0,(m+1)} - S_S^{0,(m)}|, |S_S^{1,(m+1)} - S_S^{1,(m)}|, |U_S^{(m+1)} - U_S^{(m)}|\} < \text{tol\_inner}$ .

**Step 3. Update cutoffs (outer update).**

- **Step 3.1. Separation cutoff  $p_S^*$ .** Scan forward from the previous cutoff index  $j_0^{\text{prev}}$  for the first grid point where  $S_S^{\text{max},(m+1)}(x, p_j) \geq 0$ :  $p_S^{*,\text{prop}}(x) = p_{j^*}$ . Apply damped update with damping coefficient  $\omega \in (0, 1)$ :  $p_S^{*,(n+1)}(x) = \omega p_S^{*,\text{prop}}(x) + (1 - \omega) p_S^{*,(n)}(x)$ .
- **Step 3.2. OJS cutoff  $p_S^{\text{oj}}$ .** Scan forward from  $j_0$  for the first grid point where  $S_S^{1,(m+1)}(x, p_j) - S_S^{0,(m+1)}(x, p_j) \leq 0$ :  $p_S^{\text{oj},(n+1)}(x) = p_{j^{\text{oj}}}$ . Apply the same damped update:  $p_S^{\text{oj},(n+1)}(x) = \omega p_S^{\text{oj},\text{prop}} + (1 - \omega) p_S^{\text{oj},(n)}(x)$ . The OJS policy follows as  $s^*(x, p) = \mathbf{1}\{p < p_S^{\text{oj},(n+1)}(x)\}$ .

**Step 4. Update stationary distributions (composition block).** Given  $(\theta_S^{(n)}, p_S^{*,(n+1)}, p_S^{\text{oj},(n+1)})$ , solve the stationary flow-balance system (34)–(35) (on the discretised  $p$  grid) subject to the accounting identity for each  $x$ . From the solution, compute the seeker density

$$\tilde{u}_S(x) = u_S(x) + \int_0^1 s^*(x, p) e_S(x, p) dp, \quad \tilde{\mathcal{U}}_S = \int_0^1 \tilde{u}_S(x) dx.$$

**Step 5. Update tightness by free entry (outer update).** Compute  $\mathcal{J}_S$  from (32) using the updated  $(u_S, e_S, s^*)$  and the converged  $J_S(x, p)$ . Update  $\theta_S$  using free entry (33):

$$q_S(\theta_S^{(n+1)}) = \frac{k_S}{\mathcal{J}_S}.$$

Invert  $q_S(\theta) = \mu_S \theta^{-\eta_S}$  to obtain  $\theta_S^{(n+1)}$ . Use Anderson acceleration when updating tightness, to reduce fluctuations.

**Step 6. Convergence (outer loop).** Stop if

$$\max\left\{|\theta_S^{(n+1)} - \theta_S^{(n)}|, \|p_S^{*,(n+1)}(\cdot) - p_S^{*,(n)}(\cdot)\|_\infty, \|p_S^{\text{oj},(n+1)}(\cdot) - p_S^{\text{oj},(n)}(\cdot)\|_\infty\right\} < \text{tol}.$$

Otherwise set  $n \leftarrow n + 1$  and return to Step 2.

**Link to the unskilled block.** On convergence, the skilled unemployment value  $U_S(x)$  feeds back into the training value equation (3), while the trained density  $m_S(\cdot)$  is taken as given from training completion in the unskilled block. A global equilibrium is obtained by iterating between the two blocks until these link variables are jointly consistent.

## 10 Full model equilibrium and global algorithm

A *full equilibrium* consists of unskilled-block and skilled-block objects such that:

- (F1) The unskilled block satisfies conditions (E1)–(E5) given the skilled unemployment value  $U_S(\cdot)$ .
- (F2) The skilled block satisfies conditions (S1)–(S7) given the trained population density  $m_S(x) = \frac{\phi}{\nu} t(x)$ .
- (F3) The link variables are consistent:  $U_S(\cdot)$  used in the unskilled block equals the skilled-block solution, and  $m_S(\cdot)$  used in the skilled block equals training completion from the unskilled block.

### 10.1 Global fixed-point algorithm

The full model is solved via an outer fixed point over the link variables  $(U_S(\cdot), m_S(\cdot))$ :

**Step 1. Initialise.** Choose initial guesses  $U_S^{(0)}(\cdot)$  and  $m_S^{(0)}(\cdot)$ .

**Step 2. Solve the unskilled block.** Given  $U_S^{(n)}(\cdot)$ , solve the unskilled equilibrium as in Section 7. Obtain  $(t^{(n)}(\cdot), u_U^{(n)}(\cdot), \theta_U^{(n)}, \dots)$ . Update the implied trained mass:  $m_S^{\text{new}}(x) = \frac{\phi}{\nu} t^{(n)}(x)$ .

**Step 3. Solve the skilled block.** Given  $m_S^{\text{new}}(\cdot)$ , solve the skilled equilibrium using the nested algorithm described in Section 9. Obtain  $U_S^{\text{new}}(\cdot)$  and  $\theta_S^{(n)}$ .

**Step 4. Update link variables.** Update  $U_S$  and  $m_S$ .

**Step 5. Convergence.** Stop if

$$\max\{\|U_S^{(n+1)} - U_S^{(n)}\|_\infty, \|m_S^{(n+1)} - m_S^{(n)}\|_\infty\} < \text{tol}.$$

Otherwise set  $n \leftarrow n + 1$  and return to Step 2.

## 11 Transition dynamics: evolution of densities after a parameter change

Consider an unexpected, permanent change in some parameters at calendar time  $T$ . Where the distinction matters for the initial condition or the asymptotic steady state, we denote pre-switch quantities with superscript “pre” and post-switch quantities with “post”; equations evaluated for  $t > T$  use the post-switch parameter values. At  $t = T^+$ , cross-sectional densities are predetermined by the pre-switch history. For  $t > T$ , all hazards and policies are those implied by the post-switch parameters.

**Equilibrium object.** A transition equilibrium is a collection of time paths for value functions, policies, market tightness, and cross-sectional distributions such that:

1. Value functions solve time-dependent HJB equations given  $(\theta_U(t), \theta_S(t))$ ;
2. Policies  $(\tau, p_U^*, p_S^*, p_S^{\text{oj}})$  are optimal pointwise;
3. Distributions solve the laws of motion forward in time;
4. Market tightness satisfies free entry at each date;
5. Initial distributions match the pre-switch steady state and the economy converges to the post-switch steady state.

### 11.1 Time-dependent objects and hazards

All objects are time-dependent for  $t > T$  and are evaluated under the post-switch parameter values. Let:

- $u_U(x, t)$ : untrained unemployed,
- $t(x, t)$ : workers in training,
- $u_S(x, t)$ : skilled unemployed,
- $e_S(x, p, t)$ : skilled employment by quality.

Market tightness is time-dependent:

$$\theta_U(t), \quad \theta_S(t),$$

with job-finding rates

$$f_U(t) = \theta_U(t)q_U(\theta_U(t)), \quad f_S(t) = \theta_S(t)q_S(\theta_S(t)).$$

Policies are also time-dependent:

$$\tau(x, t), \quad p_U^*(x, t), \quad p_S^*(x, t), \quad p_S^{\text{oj}}(x, t).$$

## Endogenous destruction hazards.

$$\delta_U(x, t) = \lambda_U G(p_U^*(x, t)), \quad \delta_S^{\text{end}}(x, t) = \lambda_S \Gamma(p_S^*(x, t)).$$

## 11.2 Segment accounting identities

Define the untrained (unskilled-segment) density by type:

$$m_U(x, t) \equiv u_U(x, t) + t(x, t) + e_U(x, t),$$

where  $e_U(x, t)$  denotes untrained unskilled employment density (not tracked as a separate state variable). Thus,  $e_U(x, t) = m_U(x, t) - u_U(x, t) - t(x, t)$ . Define the trained (skilled-segment) density by type:

$$m_S(x, t) \equiv u_S(x, t) + e_S^{\text{tot}}(x, t),$$

where  $e_S^{\text{tot}}(x, t) \equiv \int_{p_S^*(x, t)}^1 e_S(x, p, t) dp$  is total skilled employment mass of type  $x$ . Aggregate segment masses are

$$M_U(t) \equiv \int_0^1 m_U(x, t) dx, \quad M_S(t) \equiv \int_0^1 m_S(x, t) dx.$$

## 11.3 Laws of motion: unskilled segment

**Training stock.** Training receives inflow from untrained unemployment at rate  $\tau(x, t)$  and loses mass through completion at rate  $\phi$  and demographic exit at rate  $\nu$ :

$$\partial_t t(x, t) = \tau(x, t) u_U(x, t) - (\phi + \nu) t(x, t). \quad (36)$$

**Untrained unemployment.** Untrained unemployment receives inflow from demographic replacement and from endogenous job destruction out of unskilled employment. It loses mass due to job finding, training entry, and demographic exit:

$$\partial_t u_U(x, t) = \nu \ell(x) + \delta_U(x, t) e_U(x, t) - (f_U(t) + \tau(x, t) + \nu) u_U(x, t), \quad (37)$$

where  $e_U(x, t) = m_U(x, t) - u_U(x, t) - t(x, t)$ .

**Untrained-segment mass by type.** Because training completion moves workers from the unskilled to the skilled segment, the untrained mass  $m_U(x, t)$  evolves as

$$\partial_t m_U(x, t) = \nu \ell(x) - \nu m_U(x, t) - \phi t(x, t). \quad (38)$$

Training entry reshuffles mass within the unskilled segment, so it does not appear in  $\partial_t m_U$ ; completion removes mass from  $m_U$ .

## 11.4 Laws of motion: skilled segment

**Skilled unemployment.** Skilled unemployment receives inflow from training completion and from separations out of skilled employment (exogenous separations and endogenous destruction). It loses mass due to job finding into acceptable matches and demographic exit:

$$\partial_t u_S(x, t) = \phi t(x, t) + (\xi_S + \delta_S^{\text{end}}(x, t)) e_S^{\text{tot}}(x, t) - \left( \nu + f_S(t)(1 - \Gamma(p_S^*(x, t))) \right) u_S(x, t). \quad (39)$$

**Trained-segment mass by type.** The trained mass  $m_S(x, t)$  increases via training completion and decreases via demographic exit:

$$\partial_t m_S(x, t) = \phi t(x, t) - \nu m_S(x, t). \quad (40)$$

(Job-to-job moves reshuffle employment across firms and match qualities but do not change segment membership, hence do not enter  $\partial_t m_S$ .)

## 11.5 Aggregate mass dynamics

Integrating (38)–(40) over  $x \in [0, 1]$  yields the aggregate segment mass dynamics:

$$\dot{M}_U(t) = \nu \int_0^1 \ell(x) dx - \nu M_U(t) - \phi \int_0^1 t(x, t) dx, \quad (41)$$

$$\dot{M}_S(t) = \phi \int_0^1 t(x, t) dx - \nu M_S(t). \quad (42)$$

Since  $\ell(x)$  integrates to 1, then  $\dot{M}_U(t) = \nu - \nu M_U(t) - \phi \mathcal{T}(t)$ ,  $\dot{M}_S(t) = \phi \mathcal{T}(t) - \nu M_S(t)$ , where  $\mathcal{T}(t) \equiv \int_0^1 t(x, t) dx$ . Adding shows total population is constant:  $\frac{d}{dt}(M_U(t) + M_S(t)) = \nu - \nu(M_U(t) + M_S(t))$ , so if  $M_U(T) + M_S(T) = 1$  then  $M_U(t) + M_S(t) = 1$  for all  $t \geq T$ .

## 11.6 Initial condition at the switch

Let  $(u_U^{\text{pre}}(x), t^{\text{pre}}(x), u_S^{\text{pre}}(x), m_U^{\text{pre}}(x), m_S^{\text{pre}}(x))$  denote the pre-switch stationary densities. At the instant of the switch,

$$u_U(x, T^+) = u_U^{\text{pre}}(x), \quad t(x, T^+) = t^{\text{pre}}(x), \quad u_S(x, T^+) = u_S^{\text{pre}}(x),$$

and segment masses satisfy

$$m_U(x, T^+) = m_U^{\text{pre}}(x), \quad m_S(x, T^+) = m_S^{\text{pre}}(x).$$

For  $t > T$ , the system (36)–(40) evolves forward under the post-switch objects  $(\tau, p^*, p_S^*, \theta_U, \theta_S)$ .

## 11.7 Market tightness along the transition

At each date  $t$ , market tightness is determined by free entry given the current distributions and continuation values (under post-switch parameters).

**Unskilled market.**

$$q_U(\theta_U(t)) = \frac{k_U \mathcal{U}_U(t)}{\int_0^1 J_U(x, 1, t) u_U(x, t) dx},$$

where  $\mathcal{U}_U(t) = \int_0^1 u_U(x, t) dx$ .

**Skilled market.**

$$q_S(\theta_S(t)) = \frac{k_S}{\mathcal{J}_S(t)},$$

where  $\mathcal{J}_S(t)$  is the expected value of a contact with a random job seeker under the current distribution.

**Inversion.**

$$\theta_j(t) = \left( \frac{\mu_j}{q_j(t)} \right)^{1/\eta_j}, \quad j \in \{U, S\}.$$

## 11.8 Transition mapping

At each date  $t$ , the equilibrium mapping is:

$$(u_U, t, u_S, e_S)_t \implies (\theta_U, \theta_S)_t \implies \text{value functions} \implies (\tau, p_U^*, p_S^*, p_S^{\text{oj}})_t \implies \partial_t(u_U, t, u_S, e_S).$$

## 11.9 Exact transition equilibrium: numerical algorithm

The transition path is obtained as a deterministic equilibrium in which forward-looking value functions and forward-moving distributions are jointly consistent. This is a forward–backward fixed point.

**Time discretisation.** Fix a horizon  $T_{\max}$  large enough that the economy is arbitrarily close to the post-switch stationary equilibrium. Discretise time:

$$0 = t_0 < t_1 < \dots < t_N = T_{\max}.$$

**Step 0. Boundary objects.** Compute the pre-switch and post-switch stationary equilibria.

- Initial condition (at  $t = 0$ ): distributions equal the pre-switch steady state.
- Terminal condition (at  $t = T_{\max}$ ): value functions equal the post-switch steady state.

**Step 1. Initial guess.** Choose initial paths for market tightness:

$$\{\theta_U^{(0)}(t_n), \theta_S^{(0)}(t_n)\}_{n=0}^N.$$

**Step 2. Backward pass: value functions and policies.** Given  $(\theta_U^{(k)}, \theta_S^{(k)})$ , solve the time-dependent HJB system backward from  $t_N$  to  $t_0$ . At each date  $t_n$ :

- Solve for  $(U_S, T, U_U^{\text{search}}, U_U)$ .
- Solve match values  $(E_U, J_U, E_S^1, E_S^0, J_S^0, J_S^1)$ .
- Update policies:

$$\tau(x, t_n), \quad p_U^*(x, t_n), \quad p_S^*(x, t_n), \quad p_S^{\text{oj}}(x, t_n).$$

**Step 3. Forward pass: distributions.** Given policy paths, solve the laws of motion forward from  $t_0$  to  $t_N$ :

$$(u_U, t, u_S, e_S)_{t_0} \longrightarrow (u_U, t, u_S, e_S)_{t_N}.$$

**Step 4. Update tightness paths.** At each date  $t_n$ , update tightness using free entry:

$$q_U(\theta_U^{\text{new}}(t_n)) = \frac{k_U \mathcal{U}_U(t_n)}{\int_0^1 J_U(x, 1, t_n) u_U(x, t_n) dx},$$

$$q_S(\theta_S^{\text{new}}(t_n)) = \frac{k_S}{\mathcal{J}_S(t_n)}.$$

Invert  $q_j(\theta) = \mu_j \theta^{-\eta_j}$  to obtain  $\theta_j^{\text{new}}(t_n)$ .

**Step 5. Convergence.** Iterate Steps 2–4 until

$$\max_n (|\theta_U^{(k+1)}(t_n) - \theta_U^{(k)}(t_n)|, |\theta_S^{(k+1)}(t_n) - \theta_S^{(k)}(t_n)|) < \text{tol}.$$

The resulting paths constitute the transition equilibrium.

## 12 Model Properties

This section collects key qualitative properties of the model that shape the economic interpretation of the estimation results and the transition-dynamics exercises.

## 12.1 Training cutoff and the education margin

**Proposition 1** (Training cutoff). *Under the maintained functional forms ( $c'(x) < 0$ ,  $U_S$  and  $T$  continuous in  $x$ ), the net training gain*

$$\Delta_T(x) = -c(x) + T(x) - U_U^{\text{search}}(x)$$

*is increasing in  $x$ . Hence the optimal training policy is a cutoff rule:*

$$\tau(x) = \mathbf{1}\{x > \bar{x}\}, \quad \Delta_T(\bar{x}) = 0.$$

*Workers with innate type  $x > \bar{x}$  train upon entering unskilled unemployment; those with  $x \leq \bar{x}$  search in the unskilled market.*

*Intuition.* Higher-ability workers benefit from training through two channels: (i) the direct cost  $c(x) = e^c (1-x)e^{-x}$  is decreasing in  $x$ , and (ii) the return to skilled-market access, captured by  $T(x)$ , is increasing in  $x$  because skilled output  $\gamma_{P_S} x^{\gamma_{P_S}} p$  rises with ability. Note that under the skill-biased parameterisation, the return-to-ability channel is amplified by the exponent  $\gamma_{P_S}$ : the marginal product of ability in skilled production grows as  $\gamma_{P_S}^2 x^{\gamma_{P_S}-1} p$ , whereas in the unskilled market it grows linearly as  $P_U p$ . *Comparative statics.* The cutoff  $\bar{x}$  responds to parameter changes through several channels:

1. A rise in  $\gamma_{P_S}$  raises  $T(x)$  for all  $x$  (and disproportionately so for high- $x$  workers), lowering  $\bar{x}$  (more workers train).
2. A rise in  $P_U$  raises  $U_U^{\text{search}}$  through higher unskilled wages, raising  $\bar{x}$  (fewer workers train).
3. A rise in unskilled tightness  $\theta_U$  raises the unskilled job-finding rate and hence  $U_U^{\text{search}}$ , also pushing  $\bar{x}$  up.
4. A rise in skilled tightness  $\theta_S$  raises the expected return to training  $T(x)$ , pushing  $\bar{x}$  down.
5. A rise in skilled matching efficiency  $\mu_S$  raises the contact rate  $\kappa_S$  at any given tightness, raising the expected return to training and pushing  $\bar{x}$  down. A fall in  $\mu_S$  has the opposite effect.
6. A rise in the training cost coefficient  $c$  raises  $c(x)$  for all  $x$  and so raises  $\bar{x}$  (fewer workers train). Because  $c$  is regime-specific, crisis-era shifts in education incentives—scholarship availability, opportunity-cost shifts, the GI-Bill-like responses to unemployment—are absorbed into  $c$  rather than mis-attributed to other channels.

In a general-equilibrium crisis, all these channels operate simultaneously. The net effect on  $\bar{x}$  depends on *which* parameters shift and by how much—precisely what the multi-regime estimation recovers.

## 12.2 Separation thresholds

**Proposition 2** (Unskilled separation cutoff). *For each type  $x$ , there exists a unique reservation match quality  $p^*(x) \in (0, 1)$  satisfying  $J_U(x, p^*) = 0$ , with the closed-form expression*

$$p^*(x) = \frac{(r + \nu) U_U(x) - \lambda_U \int_{p^*}^1 S_U(x, p') dG(p')}{P_U x}.$$

*The cutoff is decreasing in  $x$ : higher-ability workers tolerate lower match quality because the surplus  $P_U x p$  grows faster in  $p$  for large  $x$ .*

**Proposition 3** (Skilled separation and OJS cutoffs). *For each type  $x$ :*

1. There exists a separation cutoff  $p_S^*(x)$  satisfying  $S_S(x, p_S^*) = 0$ . Matches with  $p < p_S^*$  are destroyed upon a quality shock.
2. There exists an OJS cutoff  $p_S^{\text{oj}}(x) \geq p_S^*(x)$  satisfying  $E_S^1(x, p_S^{\text{oj}}) = E_S^0(x, p_S^{\text{oj}})$ . Workers in matches with  $p < p_S^{\text{oj}}$  search on the job; those with  $p \geq p_S^{\text{oj}}$  do not.

Both cutoffs are decreasing in  $x$ .

*Intuition for the OJS cutoff.* Search on the job is optimal when the expected capital gain from climbing the job ladder exceeds the flow search cost  $\sigma_S$ . Workers in low-quality matches have more to gain from a new draw  $\tilde{p} \sim \Gamma$ , so they search; those already in high-quality matches do not. Because  $\sigma_S$  is regime-specific, a crisis-induced rise in the OJS cost compresses the OJS region (raises  $p_S^{\text{oj}}$ ) and reduces the ee rate.

### 12.3 Wage structure

The model generates three wage schedules, all linear in match quality  $p$  for fixed type  $x$ :

$$w_U(x, p) = \beta_U P_U x p + (1 - \beta_U)(r + \nu) U_U(x), \quad (43)$$

$$w_S^0(x, p) = \beta_S \gamma_{P_S} x^{\gamma_{P_S}} p + (1 - \beta_S) b_S + \beta_S(1 - \beta_S) \kappa_S \int_{p_S^*(x)}^1 S_S(x, p') d\Gamma(p'), \quad (44)$$

$$w_S^1(x, p) = \beta_S \gamma_{P_S} x^{\gamma_{P_S}} p + (1 - \beta_S)(b_S + \sigma_S) + \beta_S(1 - \beta_S) \kappa_S \int_{p_S^*(x)}^p S_S(x, p') d\Gamma(p'). \quad (45)$$

Key properties of the wage structure:

1. **Within-group dispersion.** Wage dispersion within each skill group arises from heterogeneity in both  $x$  and  $p$ . The distributions of  $x$  (governed by  $a_\ell, b_\ell$ ) and  $p$  (governed by  $\alpha_U$  or  $a_\Gamma, b_\Gamma$ ) jointly determine variance and skewness, which the SMM targets via higher-order wage moments. Under the skill-biased parameterisation, the convex transformation  $x \mapsto x^{\gamma_{P_S}}$  amplifies skilled-wage dispersion driven by  $x$  when  $\gamma_{P_S} > 1$ .
2. **Skill premium.** The log skill premium  $\mathbb{E}[\log w_S] - \mathbb{E}[\log w_U]$  depends on  $\gamma_{P_S}$  and  $P_U$ , the match-quality distributions, bargaining weights, and the composition of employed workers. It is directly targeted by the SMM objective.
3. **OJS wage premium.** Comparing (45) and (44):

$$w_S^1 - w_S^0 = -(1 - \beta_S) \sigma_S - \beta_S(1 - \beta_S) \kappa_S \int_p^1 S_S(x, p') d\Gamma(p') \leq 0.$$

Searching workers earn *less* than non-searchers at the same  $(x, p)$ : Nash bargaining shifts the search cost and the expected poaching loss onto the worker.

4. **Variation in bargaining across windows.** Because  $\beta_j$  is re-estimated in each window, the pass-through of match quality into wages can differ across regimes. This is what allows the model to accommodate empirically observed shifts in within-group wage dispersion (the variance and third-moment moments) without forcing them to come entirely from shifts in the match-quality distribution.



## 12.4 Market tightness and free entry

Market tightness in each sector is pinned down by the free-entry condition. For the unskilled market:

$$q_U(\theta_U) = \frac{k_U \mathcal{U}_U}{\int_0^1 J_U(x, 1) u_U(x) dx},$$

and for the skilled market:

$$q_S(\theta_S) \mathcal{J}_S = k_S,$$

where  $\mathcal{J}_S$  is the expected surplus from a random contact (equation (32)), which depends on the composition of job seekers (unemployed plus on-the-job searchers). Tightness responds to parameter changes through three channels: (i) direct shifts in vacancy costs  $k_j$  and productivity ( $P_U$ ,  $\gamma_{P_S}$ ) that change the surplus from a filled job; (ii) composition effects through changes in the unemployed pool  $u_j(\cdot)$ ; and (iii) shifts in the matching technology  $\mu_j, \eta_j$ , which directly re-shape the mapping from  $\theta_j$  to  $q_j$ . Channel (iii) is what generates a *Beveridge-curve shift* (an outward shift in the locus of  $u$  vs.  $v$  at fixed surplus): it is absent in a pure demand shock and is what distinguishes a supply / reallocation crisis from a demand crisis. During the *transition* after a parameter change, tightness adjusts gradually because the composition of job seekers inherits the pre-switch distribution and evolves slowly.

## 12.5 Transition dynamics: adjustment speed

After an unexpected permanent parameter change at time  $T$ , the economy adjusts toward the post-switch steady state. The speed of adjustment is governed by three rates:

1. **Demographic turnover**  $\nu$ . The fundamental “clock” of the model. New entrants arrive with density  $\ell(x)$  and make training decisions under the new parameters. At rate  $\nu$ , the pre-switch cohort is gradually replaced.
2. **Training completion**  $\phi$ . Workers in training complete at rate  $\phi$  and enter the skilled segment. The pipeline of trainees adjusts on a timescale of roughly  $1/(\phi + \nu)$ .
3. **Job creation and destruction**. The job-finding rates  $f_U$ ,  $f_S$  and separation hazards  $\delta_U$ ,  $\delta_S$  govern how quickly the employed/unemployed composition adjusts within each segment. These are fast relative to training and demography.

The aggregate skilled share  $M_S(t)$  satisfies

$$\dot{M}_S(t) = \phi \mathcal{T}(t) - \nu M_S(t),$$

where  $\mathcal{T}(t) = \int_0^1 t(x, t) dx$  is the total training stock. This is a stable linear ODE with convergence rate  $\nu$ , implying a half-life of  $\ln 2/\nu \approx 14$  months when  $\nu = 0.05$  (monthly). Skilled-share adjustment is therefore slow and persistent—a key feature distinguishing this model from standard search models without a training margin. *Demand shock vs. supply/reallocation shock.*

The nature of the crisis matters for the transition path, and the broader set of regime-specific parameters makes this distinction empirically testable rather than assumed:

- A **demand shock** (e.g. the financial crisis) primarily reduces  $P_U$  and  $\gamma_{P_S}$  and modestly raises shock arrival rates  $\lambda_j$ , increasing separation and reducing vacancy creation. The matching and bargaining technology ( $\mu_j, \eta_j, \beta_j, \sigma_S$ ) and the training cost  $c$  should be approximately stable across the base→crisis pair: the economy rides down the Beveridge curve rather than shifting to a new one. If  $P_U$  falls proportionally more than the effective skilled productivity, the relative return to training rises and  $\bar{x}$  falls—more workers train. Once the productivity shifters recover, the economy converges back toward the original steady state.

- A **supply / reallocation shock** (e.g. COVID-19) may additionally shift the matching technology  $(\mu_j, \eta_j)$ , the offer / damage distributions  $(\alpha_U, a_\Gamma, b_\Gamma)$ , and possibly the training cost coefficient  $c$ . The result is an outward shift of the Beveridge curve and a structurally different post-shock steady state. Whether  $\bar{x}$  rises or falls then depends on which markets and which technology channels are more affected by the shock, and the slow demographic / training clock makes the transition both longer and irreversible (in the sense that the economy does not return to the pre-shock steady state).

The multi-regime estimation identifies exactly which parameters change, and the transition-dynamics computation traces out the implied adjustment path—allowing a structural decomposition of the crisis into its component channels.